BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 6485 / 6486

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6485

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The number of neutrons present in $^{39}K_{19}$ is:

(A) 39 (B) 18

(C) 20 (D) 19

Q2. Which is a pseudo solid?

(A) CaF_2 (B) Glass

(C) NaCl (D) KCl

Q3. When $a \neq b \neq c$ and $\alpha = \gamma = 90°$, $\beta \neq 90°$ then it is:

(A) Monoclinic (B) Diclinic

(C) Triclinic (D) Polyclinic

Q4. Density of an ideal gas can be calculated by the formula:

(A) PV = dRT (B) PM = dPV

(C) d = RT/PM (D) d = PM/RT

Q5. One atmosphere is equal to:

(A) 760mm of Hg
(B) 1000mm of Hg
(C) 760cm of Hg
(D) 20 psi

Q6. The comparative rates at which the solutes move in paper chromatography, depend on:

(A) The size of the paper
(B) Rf values of solutes
(C) Temperature of the experiment
(D) Size of the chromatogram

Q7. The drying agent used in desiccator is:

(A) NaCl
(B) KBr
(C) $CaCl_2$
(D) $BaCl_2$

Q8. The number of moles of CO_2 which contain 8.0g oxygen:

(A) 0.25
(B) 0.50
(C) 1.0
(D) 1.50

Q9. The mass of one mole of electrons is:

(A) 1.008g
(B) 0.55mg
(C) 0.184g
(D) 1.673mg

Q10. Glucose is converted into ethanol by the enzyme ____ present in yeast.

(A) Urease
(B) Invertase
(C) Sucrose
(D) Zymase

Q11. If the salt bridge is not used between two half cells, then the voltage:

(A) Decrease rapidly
(B) Decrease slowly
(C) Does not change
(D) Drops to zero

Q12. A solution of glucose is 10% w/v. The volume in which 1g mole of it is dissolved will be:

(A) $1dm^3$
(B) $1.8dm^3$
(C) $200cm^3$
(D) $900cm^3$

Q13. pH of pure water is:

(A) 4.4
(B) 5.4
(C) 7.0
(D) 8.0

Q14. One calorie is equivalent to:

(A) 0.4184J
(B) 41.84J
(C) 4.184J
(D) 418.4J

Q15. Which element has smaller size?

(A) Na (B) K

(C) Al (D) Li

Q16. Which molecule has zero dipole moment?

(A) NH_3 (B) $CHCl_3$

(C) H_2O (D) BF_3

Q17. The number of electrons in the outermost shell of chloride (Cl^-) ion is:

(A) 17 (B) 03

(C) 01 (D) 08

Chemistry (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1113-XI124

Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) Calculate average atomic mass of neon.

(ii) Define molar volume. Give one example.

(iii) What is the function of electric field in mass spectrometer?

(iv) How crystals are dried in an oven?

(v) Write any two uses of chromatography.

(vi) Define crystallization.

(vii) Write any four properties of gases.

(viii) Convert 40°C into Kelvin scale.

(ix) Write two faulty assumptions of kinetic molecular theory.

(x) Differentiate between reversible and irreversible reactions.

(xi) State law of mass action.

(xii) State common ion effect.

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) What are dipole dipole forces? Give one example.

(ii) Name the factors which affect the London forces.

(iii) Cleavage of crystals is itself anisotropic behaviour. Explain.

(iv) Why ice occupies 9% more volume than liquid water?

(v) Why cathode rays are also called as electrons?

(vi) Write any four properties of positive rays.

(vii) Define spectrum and name any two types of spectrums.

(viii) For azimuthal quantum number, $\ell = 2$ and $\ell = 3$; calculate total values of magnetic quantum number.

(ix) Define solubility curve. Name its two types.

(x) Sum of mole fractions of a mixture is always equal to unity. Justify.

(xi) What do you mean by order of reaction? Give two examples.

(xii) What is the effect of temperature on rate of chemical reaction?

Q4. (Marks 12) Write short answers of any SIX parts.

(i) Name the factors influencing the ionization energy.

(ii) How sigma and pi bonds are formed?

(iii) Draw the structure of ethene ($CH_2=CH_2$) using sp^2 hybridization approach.

(iv) The bond angles of H_2O and NH_3 are not 109.5° like CH_4. Give reason.

(v) Define system and surroundings.

(vi) What is standard enthalpy of atomization? Give an example.

(vii) Differentiate between endothermic and exothermic reactions.

(viii) Define (a) Electrolysis (b) Oxidation state.

(ix) A salt bridge maintains electrical neutrality in the cell. Give reason.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.

GROUP – II

Chemistry (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6486

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The wavenumber of the light emitted by a certain source is $2\times10^{6}\ m^{-1}$. The wavelength of this light will be:

(A) 500nm
(B) 500m
(C) 200nm
(D) 5×10^{7}m

Q2. Ionic solids are characterized by:

(A) Low melting point
(B) High vapour pressure
(C) Good conductivity in solid state
(D) Solubility in polar solvents

Q3. In order to mention the boiling point of water at 110°C, the external pressure should be:

(A) Between 760 torr and 1200 torr
(B) Between 200 torr and 760 torr
(C) 765 torr
(D) Any value of pressure

Q4. Equal ____ of ideal gases at the same temperature and pressure contains ____ number of molecules.

(A) Masses ; Equal
(B) Volume ; Equal
(C) Moles ; Unequal
(D) Volume ; Unequal

Q5. The molar volume of CO_2 is maximum at:

(A) STP
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2 atm

Q6. Solvent extraction is particularly useful technique for the separation when the product to be separated is:

(A) Non-volatile or thermally unstable
(B) Volatile or thermally stable
(C) Non-volatile or thermally stable
(D) Volatile or thermally unstable

Q7. The most common laboratory example of solvent extraction is called:

(A) Ether extraction
(B) Distillation
(C) Sublimation
(D) Crystallization

Q8. 1 mole of glucose has ____ number of hydrogen atoms.

(A) 6×22.414 (B) $12\times6.02\times10^{23}$

(C) $6\times6.02\times10^{23}$ (D) $24\times6.02\times10^{23}$

Q9. The number of moles of CO_2 which contains 8g of oxygen:

(A) 0.25 (B) 0.50

(C) 1.0 (D) 1.50

Q10. In zero order reaction, the rate is independent of:

(A) Temperature of reaction (B) Concentration of reactants

(C) Concentration of products (D) None of these

Q11. If a strip of Cu metal is placed in a solution of $FeSO_4$:

(A) Cu will be deposited (B) Fe is precipitated out

(C) "Cu and Fe" both dissolves (D) No reaction takes place

Q12. The molal boiling point constant is the ratio of the elevation in boiling point to the:

(A) Molarity (B) Molality

(C) Mole fraction of solvent (D) Mole fraction of solute

Q13. Which combination is an acidic buffer?

(A) HCl + NaCl (B) $CH_3COOH + CH_3COONa$

(C) $NH_4OH + NH_4Cl$ (D) NaOH + NaCl

Q14. Which system is endothermic as well as spontaneous?

(A) $H_2O(l) \rightarrow H_2O(g)$ (B) $H_2O(g) \rightarrow H_2O(l)$

(C) $N_2(g)+O_2(g) \rightleftharpoons 2NO(g)$ (D) $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

Q15. Which molecule has zero dipole moment?

(A) BF_3 (B) H_2O

(C) NH_3 (D) $CHCl_3$

Q16. The bond order of Ne_2 is:

(A) Two (B) Three

(C) One (D) Zero

Q17. When 3d orbital is completely filled the entering electron goes to:

(A) 4s (B) 4p

(C) 4f (D) 5s

Chemistry (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 1114-XI124

Note: SECTION - I

Q2. (Marks 16) Write short answers of any EIGHT parts.

(i) Many reactions taking place in our surrounding involve limiting reactant. Justify with examples.

(ii) Define mole with example.

(iii) Discuss reason for low actual yield.

(iv) Describe sintered glass crucible.

(v) Discuss folding of filter paper briefly.

(vi) Give uses of chromatography.

(vii) Define effusion with one example.

(viii) Explain Boyl's law from kinetic molecular theory of gases.

(ix) Derive units of 'a' and 'b' used in van der Waals equation of real gas.

(x) How Kc is used to predict direction of reaction?

(xi) Discuss effect of pressure change on reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(xii) Define pKa and pKb

Q3. (Marks 16) Write short answers of any EIGHT parts.

(i) Complete these nuclear reactions: (a) ${}^{4}_{2}He + {}^{9}_{4}Be \rightarrow$ (b) ${}^{14}_{7}N + {}^{1}_{0}n \rightarrow$

(ii) Differentiate between continuous and line spectrum.

(iii) Calculate ionization energy of H-atom.

(iv) Give relationship between (a) Energy and Frequency (b) Frequency and wavelength

(v) What are advantages of vacuum distillation?

(vi) Evaporation is a cooling process. Explain why?

(vii) The crystals showing isomorphism mostly have the same atomic ratios. Explain the statement.

(viii) Molecular solids are relatively soft. Why?

(ix) Define upper consulate temperature.

(x) What are azeotropic mixture?

(xi) What do you mean by poisoning of a catalyst?

(xii) What do you mean by heterogeneous catalysis? Give two examples.

Q4. (Marks 12) Write short answers of any SIX parts.

(i) Why size of anion is always larger than its neutral atom?

(ii) Why second ionization energy is greater than first?

(iii) Define bond length. Give two factors affecting bond length.

(iv) Define bond order. Give its formula.

(v) Burning of candle is spontaneous process. Justify.

(vi) Define enthalpy of combustion. Give one example.

(vii) Why enthalpy of some compounds cannot be measured directly?

(viii) What is anodized aluminum? Give its use.

(ix) What is the function of salt bridge?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5.

Q6.

Q7.

Q8.

Q9.